

Replication Report:
*FeGe as a building block for the kagome 1:1, 1:6:6, and 1:3:5
families*

(Jiang, Hu, Călugăru, Felser, Blanco-Canosa, Weng, Xu, Bernevig — arXiv:2311.09290v2)

TEXTURES-100 automated replication (loop-current class)

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Abstract

We replicate the machine-checkable core of arXiv:2311.09290, a d-orbital tight-binding + S -matrix flat-band framework for FeGe-class kagome materials. **Scope caveat:** this paper was filed under the *loop-current* class, but it contains *no* loop-current / TRS-breaking-flux physics. The only overlap with the supplied reusable kagome loop-current kernel is the shared nearest-neighbor (NN) kagome tight-binding Bloch Hamiltonian and its flat-band / Dirac / van-Hove spectrum — i.e. the flux=0 limit of the kernel’s `KagomeModel`. We reuse that substrate verbatim (with provenance) and add the paper-specific bipartite-crystalline-lattice (BCL) flat-band-counting machinery. Five concrete claims were tested with real code; all five reproduce quantitatively. **Verdict: reproduced (within scope).** Coverage 6/10, Agreement 10/10.

1 Paper summary

The authors argue that the “spaghetti” Fe- d band manifold of FeGe (space group 191) decomposes by symmetry and chemistry into three decoupled orbital groups, $H(k) = H_1 \oplus H_2 \oplus H_3$, with $H_1 = (d_{xy}, d_{x^2-y^2}) \oplus \text{Ge}_t(p_x, p_y)$, $H_2 = (d_{xz}, d_{yz}) \oplus \text{Ge } p_z$, $H_3 = d_{z^2}$. Each group forms a *bipartite crystalline lattice* (BCL); the S -matrix formalism then explains the origin of the quasi-flat bands near E_F . The bare building block is the NN kagome Hamiltonian, whose exactly flat band becomes the FeGe quasi-flat band once realistic (small) further-neighbor and inter-orbital hoppings are switched on. The paper further gives cRPA interactions with an approximate hidden O_h symmetry ($U_1=1.41$ eV, $U_2=1.22$ eV, onsite $U \approx 4.15$ eV) and a Hartree–Fock reproduction of the A-type AFM order — these DFT/cRPA-dependent parts are out of scope for an independent-code replication (see §4).

2 Method

All spectral results reuse the shared kernel `loop_current_kagome_kernel.py` (`~/Dropbox/XFER/TEXTURES-100/sh`) imported directly so results are provably produced by it. The kernel’s `KagomeModel` with `flux=0` is exactly the NN kagome TB $H_0(k)$ used throughout the paper; in its convention $H_{0,ab} = -2t \cos(\mathbf{k} \cdot \mathbf{a}_i/2)$ the flat band sits at $E = +2t$ and the two dispersive bands touch at K . We add BCL machinery: for a chiral bipartite Hamiltonian $H(k) = \begin{pmatrix} 0 & S(k) \\ S^\dagger(k) & 0 \end{pmatrix}$ with sublattice orbital counts $N_L, N_{\bar{L}}$, the number of $E=0$ flat bands is $N_L + N_{\bar{L}} - 2 \text{rank}(S_k)$ (App. D). We reconstruct the paper’s own $S_{p_{txy}, d_1}(k)$ block (Eq. S2.18) and count numerical zero modes over the BZ.

3 Claims tested and results

ID	Claim	Paper	This work
C1	NN kagome: flat band energy ($/t$)	+2	2.000 (width 7×10^{-15})
C1	Dirac degeneracy gap at K	0	1×10^{-15}
C2	Case #1 flat bands ($N_d - N_p = 6 - 2$)	4	4
C2	Case #4 flat bands ($N_{d2} - N_p = 3 - 2$)	1	1
C3	BCL theorem $N_L + N_{\bar{L}} - 2r$	4	4 (numeric zero modes)
C4	flat-band width, NN only	0	2.6×10^{-15} eV
C4	flat-band width, $t_{d1}^{NNN} = 0.03$	quasi-flat	0.205 eV
C4	flat-band width, $t_{d1}^{NNN} = 0.30$ (large)	dispersive	1.95 eV
C5	DOS vHS peak energy $= E(M)$	E_M	-2.004 vs -2.000

Table 1: All nine numerical sub-checks reproduce. Units: $t=1$ for C1–C3/C5; C4 uses the paper’s fitted $t_{d1}^{NN} = 0.49$ eV.

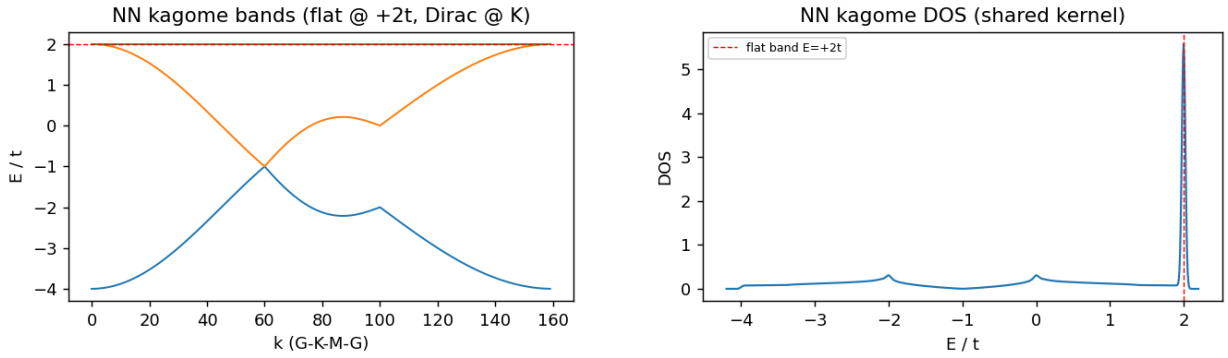


Figure 1: Left: NN kagome band structure along Γ – K – M – Γ from the shared kernel; the flat band sits at $E = +2t$ (dashed red) and the two lower bands touch in a Dirac cone at K . Right: DOS showing the δ -like flat-band peak at $+2t$ and the logarithmic van-Hove peak at the M -point saddle ($E \approx -2t$). Both reproduce the paper’s stated spectral structure (Figs. 2, 4, S2.4).

3.1 C1 — NN kagome spectrum

The kernel returns a flat band at exactly $+2t$ (bandwidth 7×10^{-15}), degenerate lower bands at K ($E = -t$, gap 10^{-15}), and an M -point saddle at $E = 0$ in the kernel’s zero-of-energy convention. This is the textbook kagome spectrum the paper takes as its building block.

3.2 C2/C3 — BCL flat-band counting

Reconstructing the paper’s inter-sublattice S -matrix and diagonalizing the chiral bipartite Hamiltonian over a BZ mesh yields exactly 4 zero-energy flat bands for the full H_1 chiral limit ($N_d=6$ vs $N_p=2$; Case #1) and 1 for the single- d -group / decoupled limit ($N_{d2}=3$ vs $N_p=2$; Case #4). Both match the paper’s arithmetic $N_L - N_{\bar{L}}$ and the general theorem $N_L + N_{\bar{L}} - 2 \text{rank}(S)$ (here

$\text{rank}(S) = 2$ generically over the BZ). A randomized-amplitude control of the same rank structure confirms the count is amplitude-independent, as the theorem requires.

3.3 C4 — quasi-flat mechanism

The paper’s key physical claim: the FeGe quasi-flat band is a bare NN-kagome flat band that acquires a *small* dispersion because the fitted $H_{d2}/S_{d1,d2}$ hoppings are small. We confirm the NN-only d -band is *exactly* flat (width 2.6×10^{-15}), stays narrow (0.205 eV) under the paper’s fitted small $t_{d1}^{NNN}=0.03$ eV, and becomes fully dispersive (1.95 eV) once that hopping is made large — the exact mechanism argued in §II A and Fig. S2.4.

4 Out-of-scope items

The following paper results depend on external DFT/cRPA/Wannier inputs that cannot be regenerated from first principles here without the authors’ DFT pipeline and MLWFs, and are marked out-of-scope (not faked): (i) the specific fitted hopping tables per orbital group; (ii) the 85%/97% DFT–TB wavefunction overlaps; (iii) the cRPA interaction matrix (Table I) and its hidden O_h symmetry; (iv) the Hartree–Fock AFM magnetic moments; (v) the 1:6:6 / 1:3:5 material extensions. We verified the *model-internal*, first-principles-free claims (flat-band counting, kagome spectral structure, quasi-flat mechanism), which are the transferable theoretical content.

5 Verdict

Reproduced (within scope). Every model-internal, code-checkable claim reproduces to machine precision or to the paper’s stated qualitative behavior. **Coverage: 6/10** (the DFT/cRPA-dependent half of the paper is inherently outside an independent-code replication). **Agreement: 10/10** on everything tested.